Task-3 C++ Project -OOP & STL(Mini application)

Goal

Build a small C++ application using classes, inheritance, STL containers and file persistence.

Requirements

- Example: Library Management, Student Grade System, Inventory Manager
- Use classes, inheritance (if applicable), and STL (vector/map/set)
- Provide a Makefile or CMake and basic unit tests

Deliverables

1. GitHub repo with code, build instructions and sample data
2. Short README explaining design decisions

Objective

To develop a Student Grade Management System using C++ with Object-Oriented Programming concepts, STL containers, and file handling.

Project Description

The project manages student records and grades using classes and vectors. Student information is stored in files for persistence.

Features

- Add Student
- Display Student Details
- Calculate Grade
- Save Data to File

Concepts Used

- Classes and Objects
- STL Vector
- File Handling
- OOP Concepts

Time Complexity

Operation	Complexity
Add Student	O(1)
Search Student	O(n)

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main.cpp

```cpp
#include <iostream>
#include <vector>
#include <fstream>

using namespace std;

class Student
{
private:
    int rollNo;
    string name;
    float marks;
    char grade;

public:
    // Function to add student details
    void addStudent()
    {
        cout << "Enter Roll Number: ";
        cin >> rollNo;

        cout << "Enter Student Name: ";
        cin >> name;

        cout << "Enter Marks: ";
        cin >> marks;

        calculateGrade();
```

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main.cpp

```cpp
        calculateGrade();
    }

    // Function to calculate grade
    void calculateGrade()
    {
        if(marks >= 90)
            grade = 'A';
        else if(marks >= 75)
            grade = 'B';
        else if(marks >= 50)
            grade = 'C';
        else
            grade = 'F';
    }

    // Function to display student details
    void displayStudent()
    {
        cout << "\nRoll Number: " << rollNo << endl;
        cout << "Name: " << name << endl;
        cout << "Marks: " << marks << endl;
        cout << "Grade: " << grade << endl;
    }

    // Function to save data into file
    void saveToFile()
```

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main.cpp

```cpp
    {
        ofstream file("students.txt", ios::app);

        file << rollNo << " "
             << name << " "
             << marks << " "
             << grade << endl;

        file.close();
    }
};

int main()
{
    vector<Student> students;

    int choice;

    do
    {
        cout << "\n===== Student Grade Management System =====\n";
        cout << "1. Add Student\n";
        cout << "2. Display Students\n";
        cout << "3. Exit\n";

        cout << "Enter Choice: ";
        cin >> choice;
```

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main.cpp

```cpp
        switch(choice)
        {
            case 1:
            {
                Student s;

                s.addStudent();

                s.saveToFile();

                students.push_back(s);

                cout << "Student Added Successfully!\n";

                break;
            }

            case 2:
            {
                if(students.empty())
                {
                    cout << "No student records found!\n";
                }
                else
                {
                    for(int i = 0; i < students.size(); i++)
                    {
```

```
main.cpp
                else
                {
                    for(int i = 0; i < students.size(); i++)
                    {
                        students[i].displayStudent();
                    }
                }

                break;
            }

            case 3:
            {
                cout << "Exiting Program...\n";
                break;
            }

            default:
            {
                cout << "Invalid Choice!\n";
            }
        }

    } while(choice != 3);

    return 0;
}
```

input

OUTPUT:

```
===== Student Grade Management System =====
1. Add Student
2. Display Students
3. Exit
```

```
2. Display Students
3. Exit
Enter Choice: 1
Enter Roll Number: 4
Enter Student Name: nafeez
Enter Marks: 70
Student Added Successfully!
```

Short Explanation

This project uses Object-Oriented Programming concepts in C++. A Student class is used to store student details and calculate grades. STL vector is used to store multiple student records dynamically, and file handling is used to save records into a text file permanently.

Conclusion

This project improved understanding of Object-Oriented Programming, STL containers, and file handling concepts in C++.